

Trusted AI Safety Expert: Module 3

Official Study Guide



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Introduction to TAISE

If AI is going to shape the world, we need people who know how to shape it safely.

The **Trusted AI Safety Expert (TAISE)** certificate, created by the Cloud Security Alliance (CSA) and Northeastern University, is a rigorous, research-backed program for professionals who build, manage, or audit intelligent systems. Through 10 comprehensive modules and a final certificate exam, learners gain practical skills to evaluate and mitigate real-world AI risks, apply AI safety and security controls, navigate compliance frameworks, and lead responsible AI adoption across industries. TAISE is more than a certificate—it's a commitment to advancing safe, secure, and responsible AI.

Receive a Respected Industry Credential

TAISE is co-issued by CSA, **the global leader in cloud security standards**, and Northeastern University, **a top academic institution for AI ethics and experiential learning**. This joint recognition enhances your professional credibility, providing a globally respected credential to highlight on resumes, LinkedIn, and professional profiles.

Acquire In-Demand Skills

Professionals will learn to work with frameworks such as **NIST AI RMF, CSA AICM, ISO standards**, and **MITRE ATLAS** while mastering **threat classification, data privacy, model governance, and AI-specific incident response**. These capabilities directly map to the urgent needs of enterprises adopting AI.

Go Beyond Theory

The **TAISE Prompt Library** transforms concepts into hands-on practice. It offers curated prompts and exercises that show learners how to use large language models to reinforce and apply safety principles from each module. While not an exam prep tool, it helps professionals **translate training into job-ready skills**.



Module 3: Generative AI Use Cases & Applications

AI is changing how we find and use information, making interactions more natural and personalized but also raising important questions about accuracy, privacy, and trustworthiness. In this module, you will look more in-depth at applications of generative AI, also known as genAI. In addition to making predictions, genAI can create new, original content such as through writing stories, drawing pictures, and creating music. GenAI models can even answer questions by learning patterns from existing data. You might have already interacted with generative AI through chatbots, voice assistants, or image generation tools.

In this module, you will learn about the benefits, uses, and misuses of this type of AI technology. Once you have this foundational knowledge, you will then reflect on the current state of AI, how it is being used in industry, and potential challenges associated with this technology. Many of the examples and case studies in this module highlight concerns related to bias, ethics, and responsible AI implementation within AI systems. You will also learn about AI safety and AI security and their distinctions. By the end of this module, you will be familiar with common real-world applications of genAI and the current ethical concerns related to genAI tools.



Access the [TAISE Prompt Library](#), a free collection of curated prompts designed to complement the TAISE training. This optional resource provides example prompts and techniques that show how to use large language models (LLMs) to explore, reinforce, and apply key ideas from each module.

Learning Objectives

In this module, you will learn to:

- Explore real-world applications of generative AI in healthcare and cyber security
- Define specialized applications including AI agents, chatbots, and multimodal systems
- Reflect on the ethical implications related to deepfakes, misinformation, and content authenticity
- Recognize the role of safety and security in generative AI applications
- Identify and categorize the different types of bias seen in AI

3.2 Module Vocabulary

Generative AI has dramatically expanded its reach and influence across diverse applications, raising both remarkable opportunities and significant ethical and safety concerns. At its core are powerful models such as **large language models (LLMs)**, which understand and generate human-like text, and sophisticated systems such as **generative adversarial networks (GANs)** and **variational autoencoders (VAEs)**, which can create highly realistic synthetic data. These advanced tools underpin technologies such as **deepfakes**, synthetic media that blur the lines between reality and fabrication, potentially harming individuals and spreading misinformation. Further enhancing capabilities, **multimodal systems**, such as ChatGPT, integrate multiple data formats—text, images, and audio—providing more comprehensive and human-like interactions. Similarly, **AI agents** and **chatbots** offer automation and interactivity, autonomously navigating complex tasks and simulating human conversation, respectively.

However, generative AI also introduces risks like **hallucinations**, instances where AI outputs plausible yet incorrect information, highlighting the importance of robust **AI safety** practices to ensure these powerful systems operate responsibly. Addressing these challenges involves ongoing ethical inquiry and proactive implementation of governance frameworks to guide responsible innovation, thus safeguarding privacy, fairness, and security while maximizing AI's transformative potential.

Below are the vocabulary terms for this module.

An **AI agent** is a software system that can observe its environment for inputs or patterns, make decisions, and autonomously take actions to achieve specific goals. AI agents can range from simple chatbots that respond to user queries to complex systems that control robots, manage trading portfolios, or coordinate multiple tasks across different platforms.

AI safety is the field of research and practice focused on ensuring that artificial intelligence systems are beneficial and reliable and do not inadvertently cause harm to humans or society. It encompasses efforts to prevent AI systems from behaving in dangerous or unpredictable ways, ensuring they operate as intended, and addressing potential risks ranging from bias and privacy violations to more existential concerns about highly advanced AI systems. AI safety involves technical approaches such as testing and verification as well as governance and policy measures to guide the responsible development and deployment of AI technologies.

AI Safety vs. AI Security

AI safety and AI security are closely related, but they deal with different kinds of risks. AI safety is about making sure systems behave as intended, avoiding issues such as bias, hallucinations, or unpredictable actions. It is focused on preventing harm that might occur even without bad intentions. AI security, on the other hand, is about protecting systems from deliberate attacks or misuse, such as someone trying to hack a model, feed it false data, or trick it into doing something harmful.

AI safety and security often overlap. For example, if an AI chatbot at a hospital accidentally reveals private patient information, it could be both a safety issue (medical information was released) and a security issue (confidential data was exposed to an unauthorized individual). In real-world settings, attackers can take advantage of safety gaps to cause security problems. That's why building trustworthy AI requires a mix of good design, strong protections, and ongoing monitoring throughout the system's life cycle.

A **chatbot** is a computer program designed to simulate conversation with human users through text or voice interactions. Chatbots can range from simple rule-based systems that follow predetermined scripts to sophisticated AI-powered systems that use natural language processing (NLP) to understand context and provide more dynamic, human-like conversations. They are commonly used for customer service, information retrieval, and automated assistance across websites, messaging apps, and other digital platforms.

Data Bias and Algorithmic Fairness

When we talk about fairness in AI, we're really asking: Is the system treating everyone equally and justly? Two key ideas here are **data bias** and **algorithmic fairness**. **Data bias** happens when flaws are introduced due to collection methods, analysis processes, or biases in human interpretation, leading to the information used to train an AI model not fully or accurately representing the real world. For example, if a healthcare AI is trained mostly on data from one demographic group, it might not work as well for others. **Algorithmic fairness** is about reducing bias, including data-related bias and proxy data bias, among other types of bias that will be covered in this module. Pursuing algorithmic fairness is vital because AI models make decisions that can affect real people—such as recommending who gets a loan, a job interview, or medical care. Understanding and addressing bias and fairness helps us build AI systems that are more ethical, accurate, and trustworthy.

Deepfakes are synthetic media created using artificial intelligence, specifically, deep learning techniques. These technologies can manipulate or generate visual and audio content with a high potential for deception, making it appear that individuals said or did things they never actually said or did. Deepfakes have raised significant concerns about disinformation (false information intended to mislead others), misinformation (false or inaccurate information), privacy violations, financial fraud, and potential harm to individuals' reputations, prompting ongoing discussions about detection methods and ethical guidelines for synthetic media creation.

Ethics is the branch of philosophy that deals with questions of right and wrong, good and bad, and moral responsibility. It involves the systematic study of principles that guide human behavior and decision-making, helping us determine what actions are morally acceptable or unacceptable. Ethics encompasses both the theoretical frameworks we use to evaluate moral choices (such as considering consequences, duties, or virtues) and the practical application of these principles in real-world situations. The development and deployment of AI systems raises fundamental ethical questions about fairness, accountability, privacy, and the potential societal impacts of automated decision-making, requiring careful consideration of how these technologies should be designed and governed to align with human values.

In the context of LLMs and generative AI, a **hallucination** refers to when a model produces information that is factually incorrect, fabricated, or not supported by its training data or the given input. These hallucinations are often presented to the user with high confidence, making them hard to detect. Hallucinations happen because these LLM models are designed to generate coherent, contextually appropriate responses based on patterns in their training data, but they don't have a built-in mechanism to verify the accuracy of what they produce. While base models do not have built-in fact-checking, certain implementations integrate external verification methods or retrieval-augmented generation (RAG) for accuracy.

A **large language model (LLM)** is a type of AI system trained on vast amounts of text data so it can understand and generate human language. These models use deep learning techniques to learn patterns, relationships, and structures in language, enabling them to perform a wide range of tasks, such as answering questions, writing text, translating languages, summarizing content, and engaging in conversations.

A **multimodal system** is a type of artificial intelligence that can process, interpret, and generate information across multiple types of data formats or "modalities" simultaneously—such as text, images, audio, video, and numerical data. For example, GPT-4 and GPT-5 are multimodal systems. Unlike single-modal systems that specialize in just one data type, multimodal systems can understand relationships between different modalities, enabling more comprehensive analysis and more natural human-computer interaction. For example, a multimodal system in a healthcare setting might analyze a radiology image, extract context from a doctor's notes, and compare lab results to assist with diagnosis. In information security, multimodal AI could combine video surveillance, text logs, and audio recordings to detect physical access violations aligned with suspicious digital activity.

Responsible AI refers to the development and deployment of artificial intelligence systems in a way that is ethical, transparent, accountable, and aligned with human values. It emphasizes fairness, privacy, safety, and inclusivity while actively working to prevent harm such as bias, discrimination, misinformation, or misuse. Responsible AI practices include using explainable models, involving diverse stakeholders in design processes, and implementing governance frameworks that ensure compliance with legal and societal standards. In enterprise and public-sector settings, responsible AI is not just a matter of ethics but a strategic imperative that builds trust, reduces risk, and ensures that AI technologies deliver value without compromising individual rights or societal well-being.

3.3 Everyday Applications of Generative AI

How is Generative AI Changing the Way We Access Information?

Generative AI (genAI) refers to a category of artificial intelligence designed to create new content by recognizing and leveraging patterns within existing data. GenAI techniques produce entirely new outputs, including text, images, audio, and video, unlike traditional AI methods that rely on pre-defined rules and algorithms to perform specific tasks. GenAI is profoundly transforming how we access, interpret, and utilize information. GenAI in the case of LLMs, for instance, allows users to interact naturally through conversational interfaces, obtaining tailored and contextually relevant responses. This shift is enabling more intuitive information retrieval, synthesis of complex topics, and even automatic generation of insights from vast datasets. However, it also introduces critical challenges, including misinformation, data privacy concerns, and algorithmic bias, underscoring the need for responsible use and vigilant oversight in professional, educational, and personal contexts.

Despite being an emerging technology, generative AI has already become part of our everyday lives. This rapid advancement underscores the importance of proactive risk assessment and ongoing monitoring as AI systems become more embedded within critical infrastructure and daily operations ([International AI Safety Report 2025](#)). In this lesson, you will explore the various ways it is currently being used, both in the business world and beyond. As you read, consider some of the ethical considerations surrounding these technologies. How do these considerations affect genAI's application and use?

How Generative AI is Changing the Ways We Access Information

Even though the technology is still emerging, generative AI has already been integrated into our daily lives. This lesson explores some of the ways it's used today, both in and outside the business realm, and discusses some of the ethical implications of both the technologies and their use cases.

Large Language Models

The generative AI systems that have gotten the most attention are large language models, such as OpenAI's GPT-4, Google's Gemini, Microsoft's Copilot, Meta's Llama 2, and many others that are designed to generate human-like text output based on the input that they get.

They can be used to assist with a variety of tasks, like creating content, answering questions, and writing code. It's a technology that many businesses are eager to adopt because it's advertised as a way to solve tasks efficiently and fast at a low cost.

However, large language models are not, as many users seem to believe, trained on the entire web, nor are they necessarily able to perform live web searches, which means that they do not have access to all the information that's currently available online.

Integration with Search Engines

Large language models can be embedded into search engines, as we've seen in projects like Google Search Generative Experience and Microsoft's Copilot. This allows them to retrieve information online in real time.

These generative AI tools are still new, and we'll likely see many more combinations with other existing tools in the near future, which means that we'll most likely come to interact with them even more, whether we know it or not.

Chatbots and Customer Interaction

These technologies have made possible a new generation of chatbots that are far better at mimicking human interaction and performing tasks than their predecessors. These chatbots are used in commercial, private, and public settings.

One example of public use is New York City's chatbot MyCity, which is a bot that helps citizens answer questions about starting and operating a business in New York City. One obvious upside to this is that it allows New York to assist citizens in 10 different languages, which makes public information more accessible for those whose first language isn't English.

The same type of chatbots have quickly become standard in large parts of the e-commerce industry, where they're often the first contact point between customers and customer service.

Benefits for Businesses

For businesses, some of the upsides include:

- Personalizing their offers
- Providing customer service 24/7
- Cutting customer service costs
- Collecting vast amounts of information on consumer behavior and preferences, which can be traded, used to target customers more efficiently, and gain a more thorough understanding of their problems

Image, Video, and Audio Generation

Generative AI is also being used to generate image and video content, as well as sound and music content. Examples of such technologies are many, and we'll just look at a few here:

- DALL-E
- Midjourney
- Imagine

These are text-to-image generator models that allow users to generate images from text prompts.

Other Applications

These are some of the most obvious uses of generative AI, but the technology is also used in a number of other sometimes less transparent ways. For example:

- Fraud detection
- Healthcare technologies
- Video games
- Coding

Looking Forward

Because the technology is just emerging, we'll most likely see people use it in many other ways in the very near future, just as we'll probably see it combined with and integrated into many existing tools and platforms.

This lesson explained that most large language models (LLMs) are not trained on the entire internet and are not necessarily even connected to the internet (though there are exceptions). This means that the outputs generated could contain outdated or inaccurate information. For this reason, it's imperative that users review any generative output before relying on or applying that output in practice.

3.4 Image Generation Example: Deep Fakes

What are Deepfakes?

Going back at least 30 years, software such as Adobe Photoshop allowed individuals to alter images. While image altering technology raises a number of significant legal and ethical issues, the landscape of risk associated with fake images has drastically shifted with the advent of generative AI tools capable of creating or modifying images.

Today, many of the major genAI tools employ automated image generation via a mechanism called diffusion. Briefly, these diffusion-based models are neural networks trained to de-noise images. This training approach where neural networks learn to de-noise images ends up allowing them to generate incredibly realistic photos, images, and even video. The term “deepfake” was coined to describe the use of “deep” neural networks to create intentionally misleading photos, images, and videos.

When prompted to create “two cyber security professionals discussing AI ethics in a futuristic digital workspace while wearing cutting-edge VR headsets,” Adobe Firefly came up with the following images:



While you can probably tell this is an AI generated image, when these tools are instead used to subtly modify existing images or video, the results can be essentially impossible to detect. In 2021, deepfakes made global news when a documentary about the deceased writer and TV star Anthony Bourdain made headlines for using genAI to create a voiceover that sounded like Bourdain had recorded it himself. The U.S. Department of Homeland Security has also highlighted the risk of deepfakes associated with evidence fabrication, corporate social engineering attacks, stock manipulation, cyberbullying, and many other crimes.

The technology underlying deepfakes is improving rapidly month by month. As a result, governments, companies, and individuals must implement increasingly sophisticated approaches for trying to detect deepfakes and also mitigate associated damages. Our legal and scientific institutions must also adapt to increasing risks associated with intentionally falsified evidence. Deepfakes will likely be an ever-growing threat in this new genAI era that we have entered.



Citations:

- [The Ethics of a Deepfake Anthony Bourdain Voice \(The New Yorker\)](#)
- [Increasing Threat of Deepfake Identities \(Department of Homeland Security\)](#)

3.5 Risks and Challenges of AI

Generative AI Use: Challenges and Considerations

In this lesson Dr. Haaber Ihle discusses some of the ethical implications and social challenges of generative AI. While these technologies do offer many benefits, it's crucial to understand the context-specific risks they pose. As you read this section,, reflect on the impact of generative AI use in creative projects. Do you see generative AI as a threat, collaborator, or tool?

How is Gen AI Being Used for Creative Endeavors

While generative AI appears to provide many benefits, there are also serious ethical implications and social challenges that come with these technologies. If we're to make sure that we build technologies that are beneficial to humans, we have to pay attention to the ethical risks that each technology and application area poses.

It's important to stress that because generative AI is so broadly applied in practice, the total sum of ethical risks can't be presented as a generic list. Each model and application area produces context-specific risks and benefits, and they must be examined individually. In other words, a generic ethics checklist is not enough. To give us an overview of the scope of ethics risks, we'll focus on the most prominent and broadly applicable ones here.

Environmental Costs

We're going to start with something that's often at the bottom of the list when we consider the ethical implications of technologies, namely the **environmental costs** of generative AI.

As they are today, large language models use high amounts of energy. While it's obvious that this poses a risk to all of us and should be taken seriously, it also poses a business risk as consumers are increasingly aware of environmental costs in business.

Undermining Truth and Truth-Telling

The second type of ethics risk has to do with the ways in which these technologies can undermine truth and truth-telling, first and foremost in society at large, but also in the culture of a single company.

One could say that this problem stems from the difference between how generative AI can be used and how it should be used. The large language models that are at the foundation of conversational AI can generate content, but they can't tell the difference between what's true and what's false. They also have no access to current information about the world as they're trained on a limited data set. Even if they do not know the answer to a question, they often simply just make stuff up.

This is often called a **hallucination**, but a more precise description would be to simply say that they make **errors** or **mistakes**. In fact, because of the stochastic nature of many of these models, you can frequently get different answers when asking the same exact question.

One reason this is ethically relevant is that we do rely on the information we have when we make decisions about our lives. As we choose from seemingly available options and their expected consequences, truthful information is critical for our agency.

Bias and Discrimination

The next type of ethical risk is that these models have been shown to be **biased and discriminatory**. For example, a recent study done by UNESCO shows how different large language models describe different categories in either positive, negative, or neutral terms. The results show that, for example, the category "gay" is more often than not described in negative terms by all of the tested models.

Similarly, there are many examples of models systematically ascribing negative traits to particular religious groups and races and positive to others.

Transparency, Explainability, and Redress

Another set of ethical risks has to do with concerns around **transparency, explainability, and redress**. These technologies offer no explanations as to how they come to offer the output that they do, nor is that process transparent to the users.

If you ask for a list of the most influential filmmakers of the last 10 years, you won't be able to tell what criteria were used to include or exclude someone from that list, nor will you necessarily get any sources or references. Why is one director included and another one excluded? We can't tell and we can't ask for any explanation.

In some ways, one could say that users are required to really leave their critical sense behind because asking questions like "how do you know?" simply no longer makes sense in this setting, especially because most of these technologies also lack a path for redress. There is nowhere to complain and no one to ask apart from the bot itself, which we've just seen can be pretty unreliable.

Artificial Empathy

The next type of ethics risk revolves around the fact that although these technologies are obviously unable to be empathetic, they're very capable of faking empathy and emotional engagement. I like to call it **artificial empathy**, and we should pay more attention to it.

Bots obviously don't have feelings, so why would they say things like "you make me laugh" or "that surprised me" or "I understand that could make you feel sad" when they are neither able to laugh, be surprised, or understand? Basically, because emotions are engaging—they make users want to stick around and interact more. Or as the Chatbot Pi puts it, "I use anthropomorphic language to build rapport and trust with the user."

But the fact is that building up such rapport can also lead to emotional dependence and over-dependence. In the worst case, this can be used as a tool for manipulation.

Privacy Concerns

Another risk is the fact that these technologies all have the potential to collect large amounts of personal data, which create **privacy concerns**. In the case of, for example, Snapchat's My AI, a mandatory chatbot often targeted at children, it turns out that all data shared with My AI is stored indefinitely and used to improve Snap's products and personalize your experience, including ads.

This is most often the rule, not the exception, and every company and individual that buys or builds or uses AI needs to consider privacy violations as a risk.

Authorship and Responsibility

Our final type of ethics risk is about **authorship and responsibility**. Many of these new tools raise questions about authorship, which can easily turn into legal questions about copyright and copyright infringement. This can go in two directions:

1. Determining authorship of AI-generated content: Who is the author of a specific AI-generated content—whether this is text, images, video, or sound?

2. Rights of original content creators: What are the possible rights of those who produced the original content that some of these models were trained on, and the possible infringement of their rights? This also extends to questions about what happens when a model is able to mimic the style of an artist to a degree where we can't tell the difference.

It also raises the question about who's responsible for the effects of the output created by generative AI.

Key Considerations

It should be noted that whereas some of these problems are much more prominent in generative AI, most of them are relevant to all types of AI. Concerns about privacy, transparency, and fairness, for example, should always be considered regardless of the type of AI.

It should also be stressed that simply being aware of these risks is not enough. A thorough and systematic approach to mitigating them is necessary.



Disclaimer—Technical Capabilities Update

Note: AI is a rapidly evolving field with technical advances continuing to emerge. Advanced retrieval-augmented generation (RAG) systems now allow LLMs to access current information and external sources. More models now also provide source citations and real-time search integration, addressing some of the limitations discussed in the original content.



Disclaimer—AI Consciousness and Empathy Claims

Note: This video presents certain views about AI consciousness, empathy, and understanding as definitive statements. These remain contested topics in the field, and current academic discourse recognizes that our understanding of consciousness and subjective experience in AI systems continues to evolve.

It should be noted that whereas some of these problems are much more prominent to generative AI, most of them are relevant to all types of AI. Concerns about privacy, transparency, and fairness, for example, should always be considered, regardless of the type of AI. Simply being aware of these risks is not enough—a thorough and systematic approach to mitigating them is necessary.

Challenges in the Cyber Sector

GenAI tools are trained not just to generate new content but also to be able to handle a wide range of tasks. As a result, the “general purpose” nature of newer genAI significantly enhances cyber threats, enabling sophisticated automated cyber attacks such as phishing, advanced malware creation, and automated vulnerability discovery. Recent AI systems have demonstrated the capability to autonomously discover previously unknown vulnerabilities, emphasizing the need for a rapid evolution of cyber security frameworks to counteract such threats ([International AI Safety Report 2025](#)).



Key Takeaways

This lesson focused on the following challenges and ethical implications of generative AI:

- **Environmental Impact:** Large language models consume high amounts of energy, creating environmental costs that pose risks to society and to the reputation of businesses as consumers become more environmentally conscious.
- **Truth and Misinformation:** Generative models cannot distinguish between true and false information, often generate fabricated content (called “hallucinations”), and provide inconsistent answers to identical questions. This undermines truth-telling both in society and within organizations, affecting people's ability to make informed decisions.
- **Bias and Discrimination:** Studies show that some generative models exhibit systematic bias, describing certain groups in negative terms while favoring others and ascribing negative traits to particular religious and racial groups.
- **Lack of Transparency and Accountability:** Most models provide little explanation for their outputs or decision-making processes, such as criteria or references. Users cannot understand the criteria used for responses or access sources and references. Additionally, there are usually no meaningful paths for redress or complaint when issues arise.
- **Artificial Empathy and Manipulation:** AI systems mimic empathy and emotional engagement to increase user interaction. This “artificial empathy” can lead to

emotional dependence and potential manipulation.

- **Privacy Violations:** These technologies collect vast amounts of personal data, often storing it indefinitely and using it for product improvement and advertising personalization. This data has the potential to be unintentionally exposed, which could lead to information leakage or sensitive data being compromised. Local and global legislation, such as the GDPR's "Right to Erasure", gives users the option to access, delete, or restrict how their data is processed.
- **Authorship and Copyright Issues:** Questions arise about who owns AI-generated content and whether using training data violates original creators' rights. This includes concerns about models mimicking artists' styles and determining responsibility for AI-generated outputs.



Pause and Reflect

Recently there has been much talk about using ChatGPT and similar LLM-based chatbots in the public sector. Considering what you now know about the ethical risks, which risks are most concerning to you?

3.6 Bias in Data Algorithms

Where Does Bias Show Up in Data and Algorithms?

This next lesson highlights the different types of biases that occur throughout the machine learning life cycle and how these biases impact fairness. This lesson also examines these issues at both the dataset and algorithm levels, each with different types of fairness issues that can arise.

As you read through this lesson, think about the tools you use daily in your work setting. What sorts of bias do you think could be present in the data?

Bias in Data & Algorithms

When generative AI is typically deployed, we encounter different types of biases that occur throughout the machine learning lifecycle, and this impacts fairness. These types of biases are not just limited to generative AI models but can also occur across all types of AI algorithms.

Let's explore how bias can be introduced both at the data set and algorithm levels and review some of the fairness issues that can arise from this and what technical methods have been used to identify and circumvent these.

Sources of Bias in Machine Learning

There are different types of biases that can be introduced at any point in the machine learning lifecycle, including at the human, data, and algorithmic level. Many data sources are human-generated for training any AI model, including generative models. Therefore, bias can be introduced at both the point of data collection or at preparation.

Data Collection Bias

Historical Bias

Bias can be introduced at the data collection stage, where the data is used for training without any kind of preparation or preprocessing.

Imagine you have asked a model to generate pictures of a CEO. What would you expect to see? Historically, women and minorities have been underrepresented in executive positions, and using training data that is not fully representative can exacerbate these disparities in representation. A dataset reflecting only the majority historical trend would contain images of older men and not much else. This is called historical bias and is not limited to any specific society, culture, or country, as this bias can occur in any environment, even when we carefully control and sample all features and data points.

A popular example of this can be found in image generation for a system called Midjourney, which works based on prompting the model using natural language to generate an image.



Academic research has found that the model is producing biased results when being asked to generate images of people and specialized professions. While the model did show both younger and older people, images generated for those in specialized roles were mostly older men. This reinforces a number of biases around the role of women, saying that less specialized work is a woman's domain, that older people do not or cannot work in non-specialized roles, and that only older men can hold specialized roles. It also raises questions around the model portraying gender as binary and not showing more fluid expressions of gender.

Representation Bias

Bias can also be introduced after the data has been collected, when a person has to make an unconscious or conscious decision about what data points should be included for training a model.

Representation bias can arise when a new dataset is built. Here, the data samples are not representative of the population the model would serve. For example, in healthcare, there are often instances where women or minority groups are underrepresented and therefore can skew predictive models. Recent research shows that computer-aided diagnoses, or CAD systems, return lower accuracy results for Black patients compared to white patients.

Mitigation Strategies

Both of these biases could have been mitigated or reduced by not just carefully examining the dataset prior to training a new model but also rebalancing your training samples. Furthermore, it is always beneficial to involve a diverse group of people in the development process to get feedback. This is, of course, not an exhaustive list of mitigation techniques, and many more exist, especially for domain or data-type-specific use cases.

Algorithm-Level Bias

Bias can also be introduced during the algorithm design, deployment, and evaluation phase of the machine learning lifecycle.

Evaluation Bias

Evaluation bias typically occurs during model evaluation, where the model's performance may be compared to out-of-domain benchmarks or other irrelevant learning models. Once you have developed a new benchmark or algorithm, it is common practice to compare performance against existing resources.

Similar to previous examples, popular commercial gender classification benchmarks have been found to mostly contain data of light-skinned subjects. This resulted in any new learning models performing poorly on facial recognition for people of color.

Algorithmic Bias

Algorithmic bias typically occurs when researchers or practitioners make algorithmic design choices, such as selecting particular regularization or optimization techniques that impact the performance of the model on different subgroups. In this case, the dataset may have been resampled and debiased, but due to the design choices, bias was introduced.

Deployment Bias

Consider the following scenario: You have trained a model to predict clothes-shopping behavior based on previous purchases on a dataset of millennial shoppers in the US but then applied it to predict the shopping behavior of senior citizens in Malaysia. It is very easy to see how the model you have built is not being used for its intended purpose, so there is no guarantee that it would actually perform well or at all. This is called deployment bias.

An example of deployment bias happened when the criminal justice system used a new AI model to predict the likelihood of a convicted criminal to relapse based on outdated historical crime data. By deploying such models in the real world, a single score can then be used by judges to decide on sentencing. Populations that historically have been disproportionately targeted by law enforcement, such as minorities and low-income communities, are now at increased risk of receiving worse sentencing due to a higher score.

Key Considerations

This is not an inclusive list, and there are other important types of biases to consider, as any model can suffer from more than one type of bias at a given time. Biases are not mutually exclusive.

3.7 Examples in Industry

Now that you have explored the fundamental concepts and applications of generative AI and examined the various types of bias that can emerge throughout the AI life cycle, it's crucial to see how these theoretical concerns manifest in real-world applications across different industries.

The following three case studies demonstrate how the bias types you've been learning about have created significant problems in high-impact AI systems that affect people's employment opportunities, personal security, and financial access. These examples span recruitment technology, facial recognition systems, and financial services, illustrating that bias in AI is not confined to any single sector but represents a pervasive challenge requiring constant vigilance. Through Amazon's discriminatory hiring tool, facial recognition systems with dramatically different error rates across demographic groups, and Apple Card's controversial credit limit algorithms, you'll see how the bias concepts you've studied translate into real consequences for individuals and society.

Each case study will challenge you to think critically about how bias emerges, persists, and can be addressed, while highlighting the importance of the responsible AI principles and ethical considerations you've been exploring throughout this module.

Example 1: Recruitment Tool Favors Male Candidates

Types of Bias:

Historical Data Bias and Gender Bias

Background:

In 2018, Amazon developed an internal AI tool to screen job applicants. The model was trained on resumes submitted to the company over a 10-year period. Reports indicate the model learned to downgrade resumes containing the word "women" (e.g., "women's chess club captain").



Explore the Situation Here:

[Why Amazon's Automated Hiring Tool Discriminated Against Women](#)



Outcome: Amazon discontinued the tool after discovering this gender-based bias. It was never used in production, but the example became a cautionary tale for bias in training data.



Pause and Reflect:

- *What led to the emergence of gender bias in this system?*
- *Could algorithmic fairness techniques have corrected this bias?*

- *If you were tasked with redesigning the system, how would you ensure fair evaluation of applicants?*

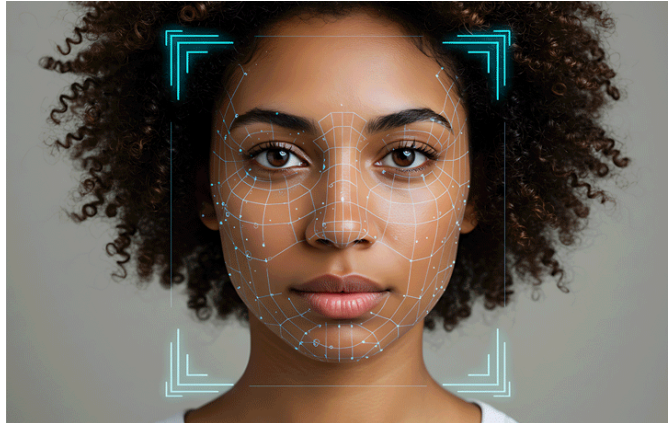
Example 2: Facial Recognition Misidentification

Types of Bias:

Demographic Bias and Sampling Bias

Background:

Studies by MIT and Stanford in 2018 showed that commercial facial recognition systems had error rates under 1% for white male faces but up to 35% for Black female faces. This was due to underrepresentation of non-white, non-male faces in training datasets.



Explore the Situation Here:

[Study finds gender and skin-type bias in commercial artificial-intelligence systems](#)

Outcome: The findings prompted major companies e.g., IBM, Microsoft, and Amazon) to pause or scale back the use of facial recognition, particularly in law enforcement.



Pause and Reflect:

- *How does biased training data affect the reliability of facial recognition systems?*
- *What are the potential social consequences of deploying such systems at scale?*
- *What measures can be taken to audit and improve dataset diversity?*

Example 3: Apple Card Allegations of Gender Bias

Types of Bias:

Proxy Data Bias and Gender Bias

Background:

In 2019, Apple and Goldman Sachs launched the Apple Card. Shortly after, multiple users, including tech entrepreneur David Heinemeier Hansson and Apple co-founder Steve Wozniak, publicly reported that they received significantly higher credit limits than their wives, despite their wives having similar or better credit histories.



Explore the Situation Here:

[Gender Bias Complaints against Apple Card Signal a Dark Side to Fintech](#)

Outcome: The controversy sparked widespread concern about algorithmic discrimination. Goldman Sachs denied using gender as an input and cited the use of complex individualized models. Critics pointed out that even if gender wasn't explicitly used in the algorithm, proxy variables like spending patterns or past credit behavior might still introduce indirect gender bias. The New York State Department of Financial Services launched an investigation into the Apple Card's credit algorithm and emphasized the need for explainability and fairness in AI-driven financial services.



Pause and Reflect:

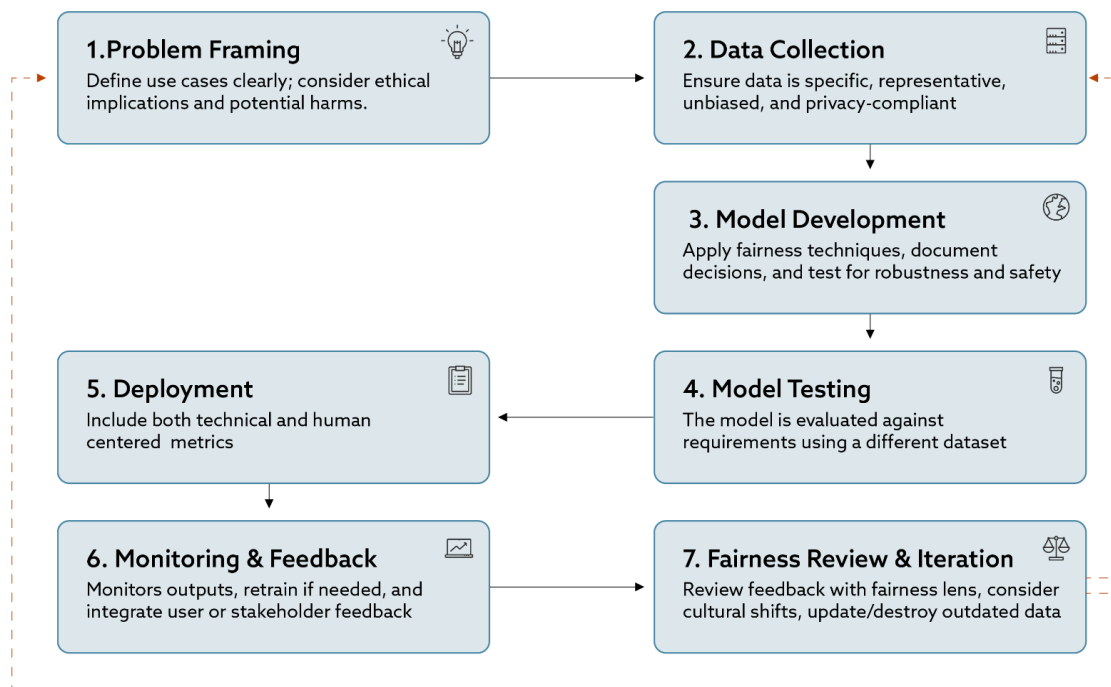
- *How can proxy variables introduce indirect bias even when sensitive attributes (such as gender) are excluded?*
- *What transparency and accountability measures should be in place for financial algorithms?*
- *How would you balance personalization in credit scoring with fairness and equal treatment?*

3.8 The Responsible AI Lifecycle

It's crucial to understand that successful AI implementation requires more than just technical capability—it demands a comprehensive responsible AI life cycle that encompasses ethical design, bias mitigation, transparency, accountability, and ongoing monitoring throughout its development and deployment.

This life cycle ensures that AI systems not only perform effectively but also maintain user safety, protect privacy, promote fairness across diverse populations, and support risk governance practices. It also supports the implementation and maintenance of risk governance (which will be covered in greater detail in later modules).

The graphic below highlights the critical steps involved in creating a responsible AI life cycle for AI applications.



Alternative text: A flow chart of the responsible AI life cycle. The steps are:

1. Problem framing, where you define your use case and consider harms and ethical implications.
2. Data collection, where you ensure data is representative, unbiased, and privacy-compliant.
3. Model development, where you apply fairness techniques, document decisions, and test for robustness and safety.
4. Model testing, where the model is evaluated against requirements using a different training set. This is followed by
5. Deployment, which involves tracking technical and human-centered metrics, and
6. Monitoring and feedback, where you continuously monitor outputs, retrain if needed, and integrate user or stakeholder feedback.
7. Phase 7 is the fairness review and iteration stage.

The case studies in the upcoming pages demonstrate how two organizations are navigating this responsible AI framework while pushing the boundaries of what's possible. We'll examine how multimodal AI systems are enabling truly personalized medicine by integrating diverse data types, as well as the emerging role of AI-powered chatbots in mental health support, examining both their potential to address critical access gaps and the unique ethical considerations they raise.

3.9 Case Study #1:

Personalized Medicine Through Multimodal AI

Background

Oncology advances have been slowed by the inability to combine heterogeneous data types in a unified way. Most machine learning models are either trained on narrow datasets (e.g., genomics only) or ignore context-rich modalities such as pathology reports or clinical narratives. This fragmentation made it difficult to fully leverage AI for personalized medicine. Memorial Sloan Kettering Cancer Center is tackling this challenge by creating an integrated, AI-ready multimodal research platform.

Explore More

- [Memorial Sloan Kettering Physician-Scientists Develop Innovative Multimodal Machine Learning Model](#)
- [MSK MIND \(Multi-Modal Integration of Data\): Projects](#)

Technical Details

MSK-MIND provides a technical and governance infrastructure to support multimodal AI research by linking:

- Radiology images and reports
- Digital pathology slides and annotations
- EHR-derived clinical data
- Patient-reported outcomes

Benefits and Outcomes

With this integrated platform, researchers can build models that span modalities, for example, correlating radiologic features with molecular signatures or predicting treatment response by fusing pathology and genomics data. Projects within MSK-MIND have developed AI systems to:

- Predict lung cancer recurrence from CT scans and histology
- Automate classification of bladder cancer subtypes from pathology
- Integrate multi-omics data to improve immunotherapy prediction

These efforts are supported by responsible data governance, patient privacy controls, and scalable cloud infrastructure for training deep learning models.

Beyond oncology, multimodal AI systems are being applied across a wide range of domains in healthcare and life sciences. These systems combine data from sources such as imaging, genomics, clinical notes, wearable sensors, and electronic health records to generate richer, more context-aware predictions. In

neurology, multimodal AI is used to correlate brain scans with cognitive assessments for early detection of Alzheimer's disease.

In cardiology, multimodal AI helps integrate echocardiograms, lab tests, and patient history to assess heart failure risk. Public health applications include monitoring disease outbreaks by combining satellite imagery, mobility data, and clinical reports. By synthesizing diverse data types, multimodal AI enables more accurate diagnostics, predictive modeling, and personalized treatment strategies across virtually every medical specialty.

AI in Medical Research

You will now read three sections from "[AI in Medical Research: Applications & Considerations](#)." This report from CSA investigates the impacts of AI in medical research in depth. It's important to note here that the report focuses primarily on general AI rather than generative AI. Despite this, the report contains ethical examples that apply to both generative AI and AI in general.

[Click here to view the assigned sections of reading.](#) We will read only a few sections of the publication, but you are encouraged to skim the report in its entirety.

Please read the following sections:

- Introduction and Abstract (pp. 5-7)
 - This section introduces key definitions related to the report, including artificial intelligence, machine learning, and deep learning. It also briefly discusses the impact of AI on drug discovery, diagnostics, and personalized medicine.
- Diagnosis and Treatment Applications (pp. 8-11)
 - This section contains examples of AI applications related to the diagnosis and treatment of conditions or diseases, such as mammography and oncology, cardiology, radiology, personalized medicine, and cognitive digital twins.
- Ethical and Legal Challenges (pp. 12-15)
 - This section discusses some ethical and legal challenges related to applications of AI in healthcare.

3.10 Case Study #2: AI Chatbots in Mental Health

AI Chatbots as Scalable Tool

Background

Mental health services worldwide face increasing demand and chronic shortages of qualified professionals. Many individuals struggle to access timely, affordable, and stigma-free mental health support. AI-powered chatbots have emerged as a scalable tool to fill some of these gaps, particularly for mild to moderate mental health concerns such as anxiety, depression, and stress.

Technical Details

- **Natural Language Processing:** Analyzes emotional context in conversations
- **Generative Responses:** Simulate empathetic, therapeutic responses based on data
- **Behavioral Tracking:** Monitors mood patterns and intervention effectiveness
- **Crisis Detection:** Identifies suicidal ideation and escalates appropriately

Benefits and Outcomes

Several companies and research groups have developed AI-powered chatbots that use natural language processing (NLP) and cognitive behavioral therapy (CBT) principles to provide support. Examples include:

- **Woebot:** An evidence-based chatbot developed by psychologists, Woebot uses daily check-ins and CBT techniques to help users manage mood and stress. A randomized controlled trial published in *JMIR Mental Health* showed significant reductions in depression symptoms after just two weeks of use.
- **Wysa:** A mental health app that combines AI chat with human coach options, Wysa adapts responses based on user input, helping individuals reflect, reframe thoughts, and access exercises grounded in positive psychology.
- **Tess:** Deployed in clinical settings and universities, Tess uses sentiment analysis and pre-programmed interventions to respond to emotional cues and support mental wellness.

Risk Considerations

While these tools can serve as first-line low-barrier support tools, triage aids, or mental health companions between therapy sessions, they are not replacements for human therapists and should not be overly relied on in crisis contexts. Users of these tools need to understand the limits and capabilities of these tools before engaging with them. These tools, like many others we have learned about in this course, require continuous human oversight to be safe and effective.



Key Takeaways

Here are the key takeaways from Woebot, Wysa, and Tess:

- Users actively engaged with the chatbots showed high satisfaction rates, with 88% of Tess conversations rated as helpful; Woebot users reported significantly higher satisfaction than the control group (4.3 vs. 3.4 out of 5).
- Process violations (inability to understand responses, repetitive conversations) were the most common complaints.
- Limitations in natural language processing can lead to user frustration.

3.11 Case Study #3: Charlotte AI

Charlotte AI: Cyber Security's First Generative AI Analyst

Background

CrowdStrike, one of the world's leading cyber security platform providers, first launched Charlotte AI in May of 2023 as an AI security analyst integrated with their Falcon XDR platform. The goal of Charlotte AI is to address the imbalance between the growth of complex cyber threats and the shortage of skilled security analysts. According to CrowdStrike, traditional security operations centers were overwhelmed by the volume and complexity of modern threats. The platform provider processes a high volume of security signals every day from their global customer base, an unprecedented threat detection training dataset for Charlotte AI.

Technical Details

CrowdStrike built Charlotte AI using an agentic workflow constructed of multiple AI agents handling discrete security functions. They leveraged AWS cloud infrastructure (Amazon Bedrock, SageMaker, OpenSearch Service) and NVIDIA NIM microservices integrations for optimal performance.

The architecture is composed of task-specific AI agents that handle data retrieval, script generation, and threat analysis. These agents search through a massive set of vector embeddings (equipped with three unique data streams: telemetry data, threat intelligence, and human-validated expert content) with a fine-tuned large language model searching for semantic similarity.

From a security perspective, Charlotte AI implements a multi-layered validation with dedicated validation agents to prevent AI hallucinations.

Benefits and Outcomes

A group of early access users found that Charlotte AI has shown a significantly faster threat analysis compared to manual methods, and qualitatively, security analysts report improved efficiency and confidence. Using the NVIDIA NIM microservices integration improved detection triage two-fold and required half the normal compute resources.

- Distributing Charlotte AI to the junior analysts at CrowdStrike allowed them to perform tasks that typically require senior expertise.
- Through automated triage of high-volume security events, Charlotte AI reduced alert fatigue.
- During the pilot deployment phase, over 5,000 security cases were successfully processed.
- Charlotte AI generated complex threat investigation of Python scripts in under 25 seconds.

Among the first few adopters, Global Data Systems and TAG cyber analysis reported that Charlotte AI added the equivalent of 20 virtual SOC analysts to their team, improved productivity by 20%, and saved over \$1 million in costs.

Risk Considerations

Many precautions were taken to avoid attacks on Charlotte AI. The hallucination prevention system mentioned earlier uses multiple validation layers and human-in-the-loop feedback mechanisms to avoid utilizing false information, leading to incorrect threat assessments or missed attacks. The system also implements role-based access controls with comprehensive audit trails, making sure the recommendations generated by Charlotte AI are appropriately authorized and trackable. For data privacy, a secure process for sensitive security information within CrowdStrike's established compliance frameworks was implemented.

However, there are always significant risks. Adversarial attacks can manipulate the AI's threat assessment capabilities, leading to false positives that overwhelm security teams, or worse, false negatives that allow real threats to pass undetected. It relies on the training data from CrowdStrike, as well as real-time telemetry and threat intelligence streams.



Key Takeaways

Here are the key takeaways from CrowdStrike's Charlotte AI:

- Multi-layered validation is essential: We should not rely on any single AI system to make autonomous security decisions without human oversight and multiple verification mechanisms.
- Domain-specific training with CrowdStrike's proprietary threat intelligence data is what made Charlotte AI so effective compared to general datasets.
- A continuous learning approach with feedback loops allows the system to enhance its capabilities but leaves it open to new threats and potential attacks.
- Data governance is a fundamental requirement: The quality and security of the training data directly impacts the reliability of Charlotte AI and any other AI system.

Sources

- [Data Poisoning: The Essential Guide | Nightfall AI Security 101](#)
- [What Are Adversarial AI Attacks on Machine Learning? - Palo Alto Networks](#)
- [Hero AI: Your Ultimate SecOps Companion](#)
- [The Golden Age of Automated Penetration Testing is Here](#)
- <https://www.exabeam.com/explainers/crowdstrike/crowdstrike-charlotte-ai-solution-overview-pros-and-cons/>
- [CrowdStrike's Charlotte AI - Enhancing productivity of Cyber Security Analysts with Generative AI built-on AWS | AWS Partner Network \(APN\) Blog](#)
- <https://blogs.microsoft.com/blog/2023/03/28/introducing-microsoft-security-copilot-empowering-defenders-at-the-speed-of-ai/>
- <https://www.crowdstrike.com/en-us/blog/crowdstrike-introduces-charlotte-ai-to-deliver-generative-ai-powered-cybersecurity/>
- [CrowdStrike Charlotte AI - Cybersecurity Excellence Awards](#)

3.12 Summary

Congratulations on completing Module 3! You have explored the transformative landscape of generative AI applications while developing critical skills in responsible AI implementation.

Key Concepts Mastered

Generative AI Fundamentals: You can now distinguish between different types of generative AI systems, including large language models (LLMs) that function without constant web connectivity and text-to-image generators like DALL-E and Midjourney. You learned that these systems create new content by recognizing patterns in existing data, fundamentally changing how we access and utilize information.

Real-World Applications: Through detailed case studies, you analyzed how generative AI is revolutionizing multiple domains:

- **Healthcare:** From AlphaFold's protein structure prediction accelerating drug discovery (reducing 10-15 year timelines by 30-50%) to multimodal AI systems at Memorial Sloan Kettering integrating radiology, pathology, and genomic data for personalized cancer treatment
- **Mental Health:** AI chatbots such as Woebot demonstrating measurable therapeutic benefits through randomized controlled trials
- **Content Creation:** The widespread adoption of AI for writing, visual art, and multimedia generation as the most prevalent current application

Specialized Systems: You evaluated advanced applications including AI agents that autonomously navigate complex tasks, conversational chatbots providing customer service and information retrieval, and multimodal systems such as ChatGPT that process text, images, and audio simultaneously for more comprehensive human-computer interaction.

Critical Insights

Bias Recognition: You identified systematic biases across the AI life cycle, from Amazon's recruitment tool discriminating against women to facial recognition systems showing 35% error rates for Black women versus under 1% for white men. You learned to categorize bias types including historical data bias, representation bias, algorithmic bias, and deployment bias, understanding that bias can be introduced at every stage, from data collection through model evaluation.

Content Authenticity Challenges: You explored the implications of deepfakes, AI hallucinations (fabricated but plausible information), and the inability of LLMs to distinguish between true and false information. The image generation exercise revealed how AI tools perpetuate stereotypes, showing limited diversity in representations of cyber security professionals.

Real-World Consequences: Through the Apple Card credit limit controversy and other industry examples, you examined how algorithmic decisions affect people's access to employment, financial services, and healthcare, understanding that bias impacts extend beyond direct users to entire

communities and society. You learned how generative AI is transforming cyber security by serving as both a powerful defensive tool and a significant new threat vector. You examined how generative AI creates new cyber security challenges, including the use of deepfakes for social engineering attacks, automated malware creation, sophisticated phishing campaigns, and the potential for adversarial attacks and data poisoning that could compromise AI-powered security systems.

Successful implementation of AI in cyber security requires robust governance frameworks, multi-layered validation systems, continuous human oversight, and comprehensive understanding of both the defensive capabilities and offensive risks that generative AI introduces to the cyber security landscape.

AI Applications in Cyber Security: You learned how generative AI is transforming cyber security by serving as both a powerful defensive tool and a significant new threat vector. You examined how generative AI creates new cyber security challenges, including the use of deepfakes for social engineering attacks, automated malware creation, sophisticated phishing campaigns, and the potential for adversarial attacks and data poisoning that could compromise AI-powered security systems.

Successful implementation of AI in cyber security requires robust governance frameworks, multi-layered validation systems, continuous human oversight, and comprehensive understanding of both the defensive capabilities and offensive risks that generative AI introduces to the cyber security landscape.

Responsible AI Framework

You were introduced to the responsible AI life cycle, understanding that successful implementation requires more than technical capability—it demands ethical design, bias mitigation, transparency, accountability, and ongoing monitoring throughout its development and deployment. This framework proved essential in the healthcare case studies, where patient safety, privacy protection, and fairness across diverse populations are paramount.

Looking Ahead: Your Continued Journey in Generative AI

The field of generative AI is evolving at unprecedented speed, with new applications, capabilities, and ethical challenges emerging regularly. Your learning in this module provides a strong foundation, but the journey continues:

Stay Current With Emerging Applications: Monitor developments in sectors where generative AI is rapidly expanding. Follow reputable AI research publications and industry reports to understand new use cases and their implications.

Develop Practical Skills: Consider pursuing hands-on experience with AI tools relevant to your field. Whether it's experimenting with prompt engineering, exploring AI-assisted coding, or understanding AI integration in your industry, practical skills will complement the foundation you have gained through this module.

Apply Your Critical Thinking: Use the ethical reasoning skills you've been refining in this course to evaluate new AI tools and applications as they emerge in your workplace and daily life. Your ability to

identify potential biases and ethical concerns is increasingly valuable as AI becomes more pervasive.

The responsible development and deployment of generative AI requires ongoing vigilance, continuous learning, and active participation from professionals like you who understand both the transformative potential and ethical complexities of these technologies. Your journey in understanding generative AI has just begun—stay curious, stay critical, and stay engaged with this rapidly evolving field that will continue to reshape our world.

Change Log

Version #	Approved by	Notes
20251015	HMR	Initial release